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日期　05/05

實驗八：Seven Segment using Verilog

題目: C108110249

一、實驗目的：

練習Verilog 設計sequential circuit, 控制 seven segment.

二、實驗步驟與結果：

請先 download the 範例 to verify the Seven Segment

1: 是否可 download 到 FPGA 板

2: Seven Segment 功能是否正常?

三、a: HDL code for the above circuit:

```
// Start of main code
//
module sevensegment(clk,rst,led, SSCom, DisOut);
input clk, rst;
output [3:0] led;
output [3:0] SSCom;
output [7:0] DisOut; // Output to seven segment
//
led          u1(clk, rst, led, SSCom);
hextoseg   u2(led, DisOut);

endmodule
//

//

module led(clk, rst, led, SSCom);
//
input     clk, rst;       // input to the led module
output   [3:0]led;       // led output with 8 bits
output   [3:0]SSCom;   // this is the com enable signal for seven segment
reg [3:0]led;   // define the led register with 8 bits
reg [3:0]SSCom;   // define the SSCom register
```

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```
reg [23:0]count;    // The count register is for dealy
reg [4:0]count_no;
reg state;
//

always @(posedge clk )       // at the rising edge
begin
      state=0;
      if(rst)    // the rst is high
      begin    // most of the time will enter this begin - end
       //////////////up////////////
           if(count_no<3 && state==0) // test for the delay

               begin
                   if(count == 24'hEB8D80)
                     begin
                    count_no = count_no+1; // clear the count, re-start
                    led = led + 1;      // shift right one bit of the led rgister


                     if(count_no==3)
                         begin
                               count=0;
                         count_no=0;
                         state=1;
                         end
                end
                    else
                    begin
                        count = count + 1;    // delay time    + 1
                    end
               end
                        //////down//////////
        if(count_no<2 && state==1) // test for the delay
            begin
                      if(count == 24'hEB8D80)
                        begin
                        count_no = count_no+1; // clear the count, re-start
```

```
                led = led - 1;     // shift right one bit of the led rgister


                  if(count_no==2)
                   begin
                         count=0;
                     count_no=0;
                     state=0;
                    end
                 end
               else
                begin
                 count = count + 1;    // delay time    + 1
                end
          end
     end
      ////////////////////////
     else
     begin                          // rst =0 , initail setting
          count = 0;
          led = 4'h0;
          SSCom = 4'hE;      // disable the SSCom
     end
end


endmodule

//
//    HEX convert to seven segment
//
//
module hextoseg(din, seven_seg);
input      [3:0]din;
output    [7:0] seven_seg;
reg [7:0] seven_seg;

//    dot h g f e d c b a
```

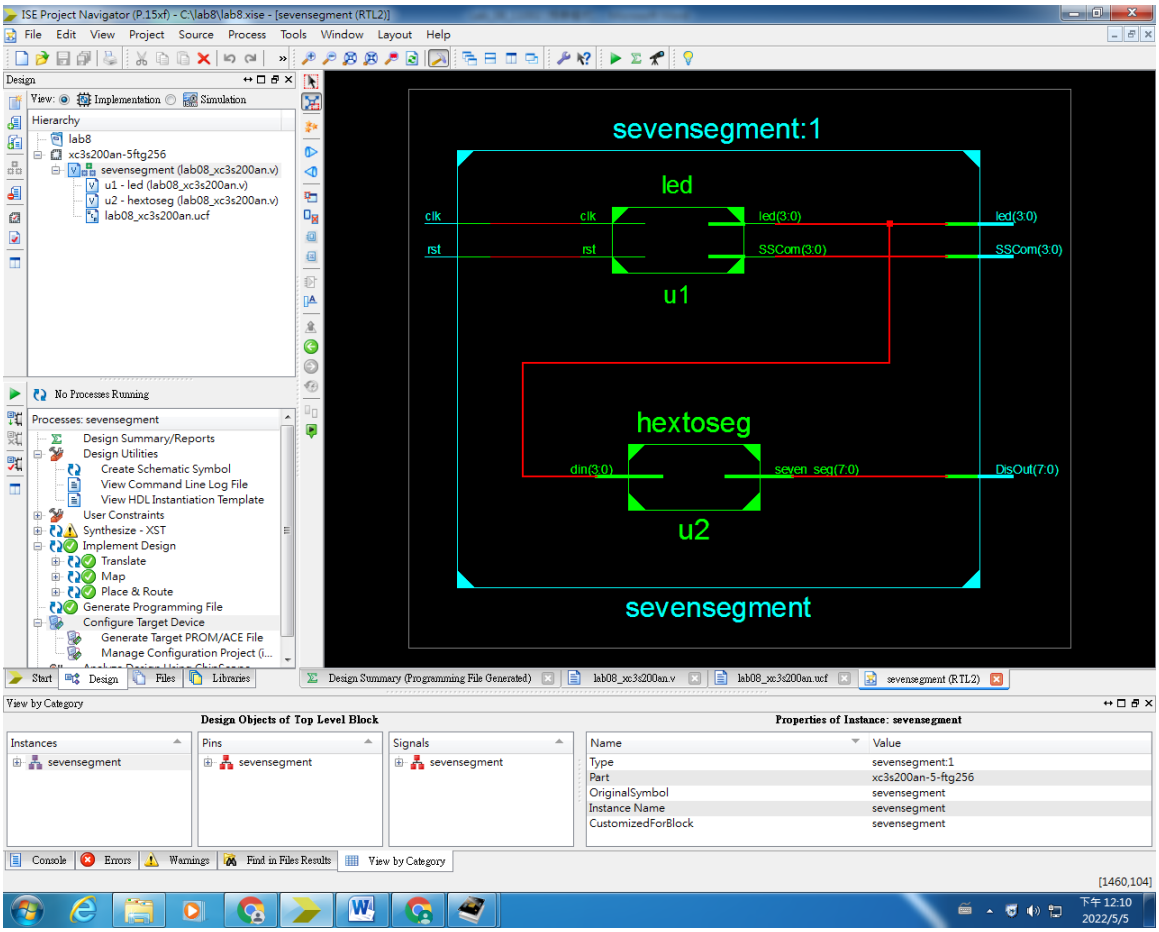
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日期　05/05

```
// low:  led on
//hexadecimal code converter seven segment code
always @(din)
begin
  case (din)
    0:  seven_seg<=8'b11000000;
    1:  seven_seg<=8'b11111001;
    2:  seven_seg<=8'b10100100;
    3:  seven_seg<=8'b10110000;
    4:  seven_seg<=8'b10011001;
    5:  seven_seg<=8'b10010010;
    6:  seven_seg<=8'b10000011;
    7:  seven_seg<=8'b11111000;
    8:  seven_seg<=8'b10000000;
    9:  seven_seg<=8'b10011000;
    10: seven_seg<=8'b10001000;
    11: seven_seg<=8'b10000011;
    12: seven_seg<=8'b11000110;
    13: seven_seg<=8'b10100001;
    14: seven_seg<=8'b10000110;
    15: seven_seg<=8'b10001110;

  endcase
end
endmodule
```

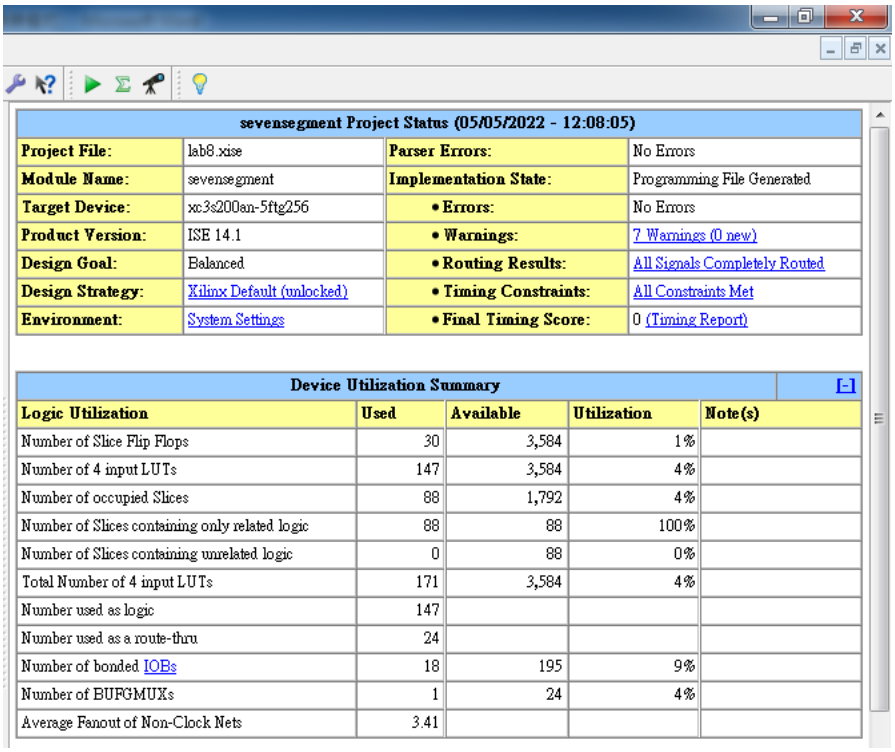
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B: ise 14.1 Synthesis的schematic：



四、ise 14.1 顯示出的 resource 使用情形

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sevensegment Project Status (05/05/2022 - 12:08:05)			
Project File:	lab8.xise	**Parser Errors:**	No Errors
Module Name:	sevensegment	**Implementation State:**	Programming File Generated
Target Device:	xc3s200an-5ftg256	• **Errors:**	No Errors
Product Version:	ISE 14.1	• **Warnings:**	7 Warnings (0 new)
Design Goal:	Balanced	• **Routing Results:**	All Signals Completely Routed
Design Strategy:	Xilinx Default (unlocked)	• **Timing Constraints:**	All Constraints Met
Environment:	System Settings	• **Final Timing Score:**	0 (Timing Report)

Device Utilization Summary				[-]
Logic Utilization	**Used**	**Available**	**Utilization**	**Note(s)**
Number of Slice Flip Flops	30	3,584	1%	
Number of 4 input LUTs	147	3,584	4%	
Number of occupied Slices	88	1,792	4%	
Number of Slices containing only related logic	88	88	100%	
Number of Slices containing unrelated logic	0	88	0%	
Total Number of 4 input LUTs	171	3,584	4%	
Number used as logic	147			
Number used as a route-thru	24			
Number of bonded IOBs	18	195	9%	
Number of BUFGMUXs	1	24	4%	
Average Fanout of Non-Clock Nets	3.41			

五、拍照 ise 14.1 七段顯示出的情形(1 picture)

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